

Remotely Mapping the World’s Cell Towers

Paper #XXX

Abstract

1 Introduction

2 Background

In this section, we describe systems that enable mobile devices to determine their own position using nearby wireless infrastructure. Then, we define and describe the identifiers used by one type of these systems.

2.1 Positioning Systems

Wireless positioning systems are used by mobile operating systems to compute their own locations. These systems use landmarks—either Wi-Fi access points (APs) or cellular base stations—with known locations to provide devices attempting to self-geolocate with reference points to do so. Wi-Fi Positioning Systems (WPSes) use Basic Service Set Identifiers (BSSIDs)—MAC addresses of Wi-Fi APs—that are broadcast in 802.11 Wi-Fi management frames to identify nearby landmarks, while Cellular Positioning Systems (CPSes) use identifiers broadcast by cellular Base Station Transceivers (BTSes) to identify these landmarks.

While prior work [3, 9, 10] examined the operation and privacy issues associated with WPSes, we exclusively study the operation and privacy issues related to CPSes in this work.

2.2 Cellular Infrastructure Identifiers

To advertise service for a cellular network, cellular BTSes advertise identifiers over the air to clients within transmission range. These identifiers indicate which Mobile Network Operator (MNO) the BTS access to via the Public Land Mobile Network (PLMN) which consists of an International Telecommunications Union (ITU)-assigned Mobile Country Code (MCC) and a nationally-administered Mobile Network Code (MNC). Cellular infrastructure identifiers also indicate a geographic area and unique identity of the BTS or BTS

sector. These two fields are assigned by MNOs to their cellular infrastructure in a manner of their choosing. When taken together, these four values should uniquely identify a specific cell for a MNO.

The name and size of the two fields assigned by an MNO vary according to the technology generation of the infrastructure they describe. For instance, in Global System for Mobile Communications (GSM) (2G) and Universal Mobile Telecommunications System (UMTS) (3G) networks, a *Cell Global Identity (CGI)* consists of a PLMN, a 16-bit regional identity called the Location Area Code (LAC), and a 16-bit Cell ID (CID). In Long Term Evolution (LTE) (4G), an *Evolved Universal Terrestrial Radio Access Network (E-UTRAN) Cell Global Identifier (ECGI)* consists of a PLMN, a 20-bit eNodeB ID, and an 8-bit CID. In New Radio (NR) (5G) networks, an *NR Cell Global Identifier (NCGI)* uniquely describes a cell, consisting of a PLMN, a 28-bit gNodeB ID, and an 8-bit CID. Table 1 summarizes the types of cellular infrastructure identifiers and their sizes by technology generation.

In this work, we amass a global corpus of infrastructure identifiers across 2-5G cellular technologies using Apple’s CPS. Because we are concerned with identifying, geolocating, and longitudinally tracking these identifiers via a CPS, rather than transmitting or receiving these identifiers with our own hardware, we are able to disregard many of the nuances in naming and bit-size between technology generations. In this work, when the distinction between the identifier used to specify a cell’s location area or region is immaterial, we refer to this identifier generally as the *location identifier*. We refer to the cell identifier, irrespective of generation and bit-length, as the *cell identifier* throughout this work.

2.3 Apple’s CPS

Lastly, we describe the operation of one CPS in particular—that run by Apple to support devices in its ecosystem. Apple’s CPS operates in two distinct but concurrent phases. In the *CPS data collection* phase, cellular Apple devices (e.g. iPhones,

Table 1: Summary of cellular network identifiers by cellular technology generation.

Technology	Generation	Unique Identifier	Network Identifier	Location Identifier	Cell Identifier
GSM	2G	Cell Global Identity	PLMN (5-6 digits)	LAC (16 bits)	Cell ID (16 bits)
UMTS	3G	Cell Global Identity	PLMN (5-6 digits)	LAC (16 bits)	Cell ID (16 bits)
LTE	4G	E-UTRAN Cell Global Identifier	PLMN (5-6 digits)	eNodeB ID (20 bits)	Cell ID (8 bits)
NR	5G	New Radio Cell Global Identifier	PLMN (5-6 digits)	gNodeB ID (28 bits)	Cell ID (8 bits)

cellular-capable iPads) collect identifiers advertised by nearby cellular BTSes. Periodically, these devices upload the identifiers they observed, along with the locations they were at when they observed them, the received signal strength of the observation, and other metadata information. These reports populate the database of cellular landmarks that Apple uses. Apple devices need not connect to a particular BTS for it to be used as a landmark, and they report cellular infrastructure for MNOs they are not subscribers of.

In the *CPS geolocation* phase, a device in Apple’s ecosystem uses these previously-collected cellular landmarks for geolocation. For an Apple device to obtain its location using the CPS, it samples the cellular frequency bands for nearby identifiers, and queries a specific Apple HTTPS API endpoint for this purpose. Apple, if it has entries for the cellular identifiers requested by the client in its database, will return the geolocations of those landmarks. Then, using the received signals strengths from those nearby cellular landmarks, the Apple device can estimate its own position. Like Apple’s WPS, Apple’s CPS also returns additional, unrequested information. Apple’s CPS provides up to 100 additional geolocated cells *nearby* one that was requested, presumably to provide as many useful landmarks to the requesting client as possible.

Apple’s CPS differentiates between BTSes for different cellular technologies by placing them in different locations in the request and response messages. Prior work [2, 4] identified that the CPS differentiates between NR (5G), LTE (4G), and a combined GSM (2G) and UMTS (3G). It also handles BTSes for Code Division Multiple Access (CDMA), CDMA Evolution-Data Optimized (EV-DO), and Synchronous Code Division Multiple Access (SCDMA). We use this knowledge to conduct the first global, large-scale survey of cellular BTSes, as well as examine the security and privacy ramifications of an adversary having the ability to accumulate this data.

3 Related Work

Several smartphone applications have been developed to attempt to detect rogue base stations. Park et al. evaluated five popular IMSI catcher applications—*SnoopSnitch* [6], *Cell Spy Catcher*, *GSM Spy Finder*, *Darshak* [11], and the *Android IMSI Catcher Detector (AIMSICD)* [1],—for Android in order to evaluate their efficacy against rogue base stations using a rogue base station framework they developed called *White-Stingray* [8]. Since this work was published in 2017, several

Table 2: OpenCellID BTS counts by cellular generation from March 2025.

Technology	Count
GSM	1,300,596
UMTS	473,432
CDMA	28
LTE	2,726,803
NR	120,009
Total	4,620,869

of these applications are no longer available in the Google Play Store.

More recently, Arnold et al., developed *CellGuard*, an iOS application for analyzing nearby cellular networks to detect possible rogue base stations [2]. *CellGuard* uses several factors, including whether or not a base station is present in Apple’s CPS, to make this determination. Unlike their work, which focuses on detecting a local adversary in real time, this work seeks to map the world’s cellular infrastructure and detect malicious and suspicious rogue base stations longitudinally across time and space. We leverage their reverse-engineering of Apple’s CPS request and response formats to conduct our study.

Most closely related to this work is Rye and Levin’s investigation of the security and privacy vulnerabilities associated with Apple’s WPS [9]. The authors demonstrated that an unprivileged attacker can amass hundreds of millions of Wi-Fi BSSID geolocations by querying Apple’s WPS with little *a priori* knowledge. We examine many of the questions, but applied to cellular infrastructure: how might one remotely discover cellular BTSes around the world using Apple’s CPS? Are there any security and privacy ramifications of providing free, unlimited access to an API that provides such information?

4 Methodology

This section details our methodology for geolocating cellular infrastructure using Apple’s CPS.

4.1 Bootstrapping BTSes

Apple’s CPS requires Apple devices to submit the unique identifiers of nearby BTSes in order to receive their geoloca-

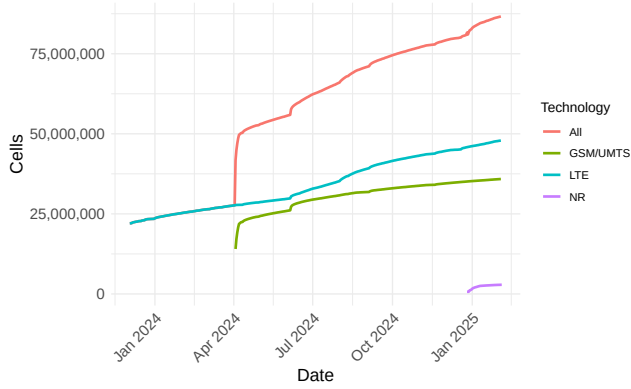


Figure 1: Cumulative number of geolocated cells over time, geolocated querying Apple’s CPS.

Table 3: Collected cellular identifier counts by cellular technology.

Technology	Count
GSM/UMTS	36,041,875
LTE	48,107,561
NR	2,891,114
Total	87,040,550

tions. However, beyond the PLMN, which identifies an MNO, there is no universal structure or format for how cells are numbered. Doing so is left up to individual MNOs, and their numbering plan may vary substantially from other MNOs, even within the same country. Further, because the combined location and cell identifier address spaces range from 28 to 32 bits, depending on the technology (1), guessing within or exhaustively enumerating the identifier space is an inefficient way to accumulate geolocated cell data.

This presents a chicken and egg problem—how might one gather a baseline of geolocated BTS data to bootstrap the geolocation? To solve this dilemma, we turned to a free database provided by OpenCellID [7]. OpenCellID provides worldwide, geolocated BTS data for free

5 Results

Using the cell identifier collection methodology outlined in §4, we accumulated over 87M unique cell identifiers by querying Apple’s CPS. Figure 1 depicts the cumulative number of cellular identifiers discovered over time.

6 Case Studies

6.1 Testing, Internal, and Non-Existent MCCs

The ITU MCC assignments reserve certain MCCs for use in network testing (MCC 001) and for use in internal or private

networks (MCC 999) [5]. Further, MCCs beginning with “1” or “8” are not assigned by the ITU. Our data includes many examples of base stations assigned one of these MCCs.

6.2 Out-of-Country MCCs

6.3 Moving Base Stations

7 Conclusion

References

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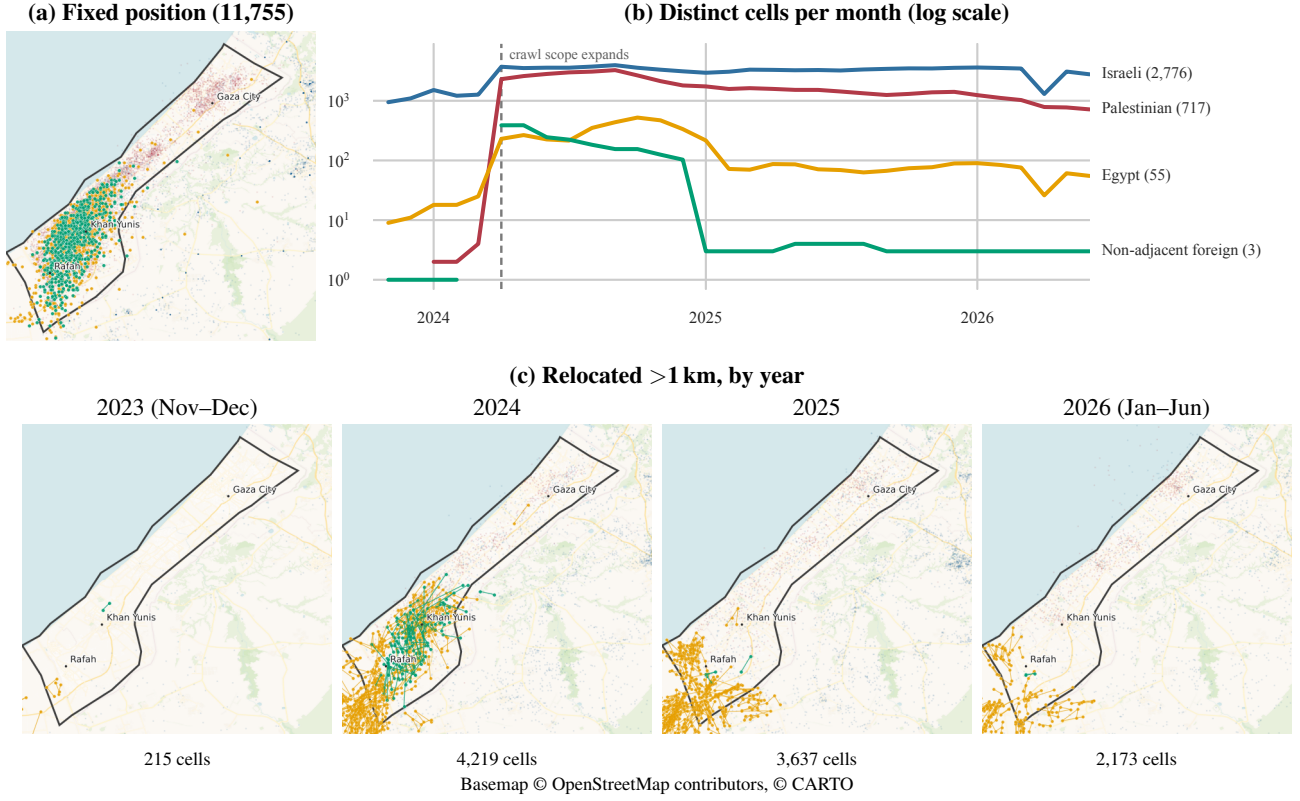


Figure 2: Foreign network identities in the Gaza Strip cluster at its southern end and are confined to 2024, while Palestinian, Israeli and Egyptian identities persist throughout.

[11] Swapnil Udar and Ravishankar Borgaonkar. Darshak. <https://github.com/darshakframework/darshak>.

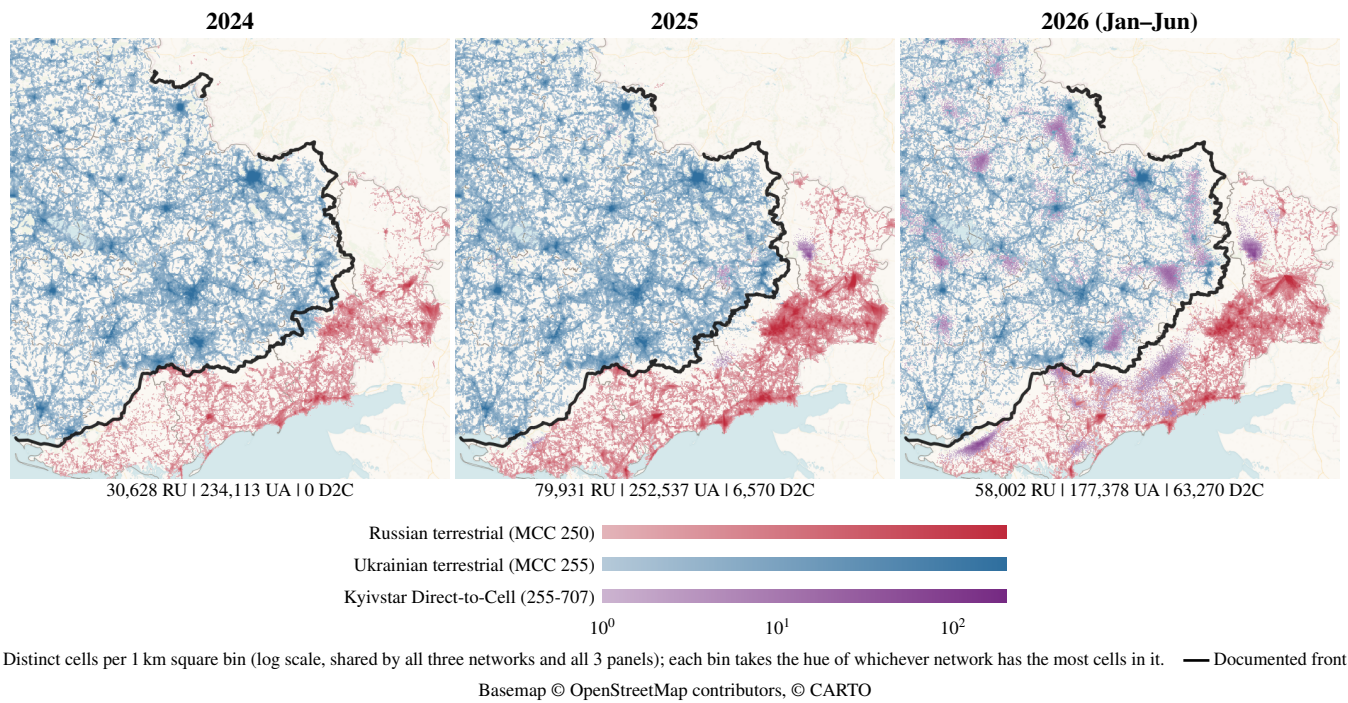


Figure 3: Density of distinct Russian, Ukrainian and Kyivstar Direct-to-Cell satellite cells across Ukraine, by year, on one shared scale, against the documented front.